Ground Effect on NACA 2412 Airfoil

Detailed Procedure, Results, and Verification

February 26, 2026

Abstract

This report studies how ground height affects lift and pressure on a NACA 2412 airfoil. A 2D vortex-collocation model with image vortices is used. The report includes: (1) C_L vs. α, (2) C_p vs. x/c in the requested format (one panel per h/c, multiple α curves inside each panel), (3) γ vs. x/c, and (4) a verification plot where model C_p is compared with XFOIL reference C_p from the published XFOIL method.

1 Objective

The goal is to quantify how changing h/c changes:

- lift coefficient C_L,
- pressure coefficient C_p,
- and vortex-sheet strength γ.

2 Detailed Procedure and Methodology

2.1 Airfoil and Cases

- Airfoil: NACA 2412.
- Chord: $c = 1$.
- Heights: $h/c \to \infty$ (no wall), 1.00, 0.50, 0.15.
- Angles:
  - C_L: $\alpha = -8^\circ$ to 12°,
  - C_p, γ: $\alpha = -4^\circ, 0^\circ, 4^\circ, 8^\circ, 12^\circ$.

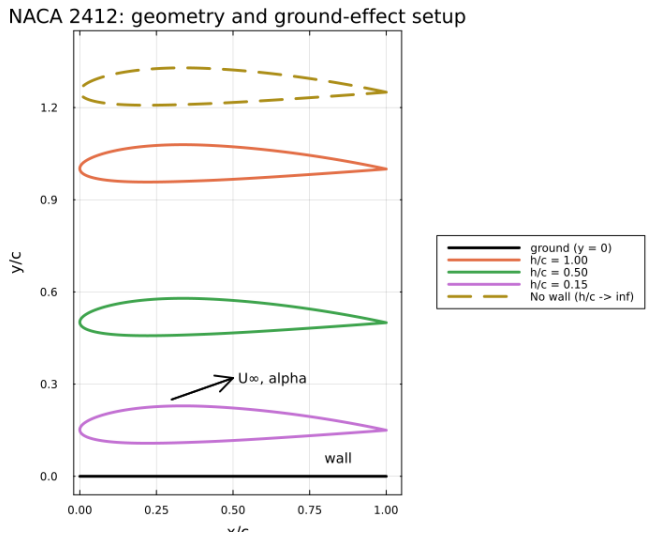

Figure 1: NACA 2412 setup diagram: airfoil placement for each h/c and ground wall at $y = 0$.

2.2 Numerical Model (step by step)

1. The NACA 2412 camber line is defined from standard 4-digit equations.
2. The chord is split into $N = 120$ panels using cosine spacing (more points near leading edge).
3. On each panel:
   - vortex point at quarter-panel location,
   - collocation point at three-quarter-panel location.
4. Normal and tangent vectors are computed at each collocation point.
5. The velocity influence of each vortex on each collocation point is assembled into a linear system.
6. Ground effect is added using image vortices across $y = 0$ with opposite sign.
7. No-penetration is enforced at all collocation points:

$$\mathbf{A}\,\boldsymbol{\Gamma} = \mathbf{b}.$$

8. The system is solved for panel strengths Γ_j.

2.3 Post-processing Equations

- Lift coefficient:

$$C_L = \frac{2\sum_j \Gamma_j}{U_\infty c}.$$

- Sheet strength:

$$\gamma_j = \frac{\Gamma_j}{\Delta s_j}.$$

- Upper-surface tangential speed:

$$V_{t,\text{upper}} = V_{t,\text{line}} - \frac{\gamma}{2}.$$

- Pressure coefficient:

$$C_p = 1 - \left(\frac{V_t}{U_\infty}\right)^2.$$

2.4 Verification Setup

To verify the C_p implementation, a no-ground case is compared against XFOIL C_p for the same airfoil and angles.

- Reference method: XFOIL (Drela, 1989).
- Verification angles: $\alpha = 0°, 4°, 8°$.
- Comparison metric: RMSE between model and XFOIL C_p on common x/c overlap.

3 Results

3.1 C_L vs α

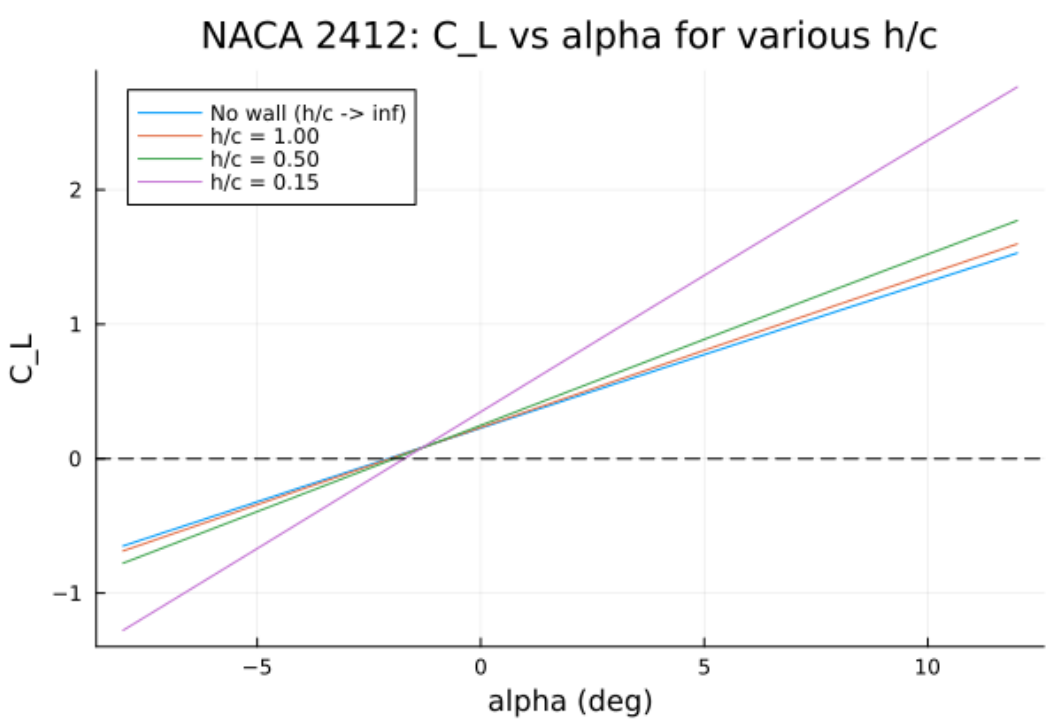

Figure 2: NACA 2412: C_L vs. α for different h/c.

Table 1: Linear fit from computed C_L curves.

Case	$dC_L/d\alpha$ (per deg)	C_L at $\alpha = 0°$
No wall ($h/c \to \infty$)	0.10911	0.22615
$h/c = 1.00$	0.11435	0.23112
$h/c = 0.50$	0.12763	0.24637
$h/c = 0.15$	0.20244	0.34628

3.2 C_p vs x/c

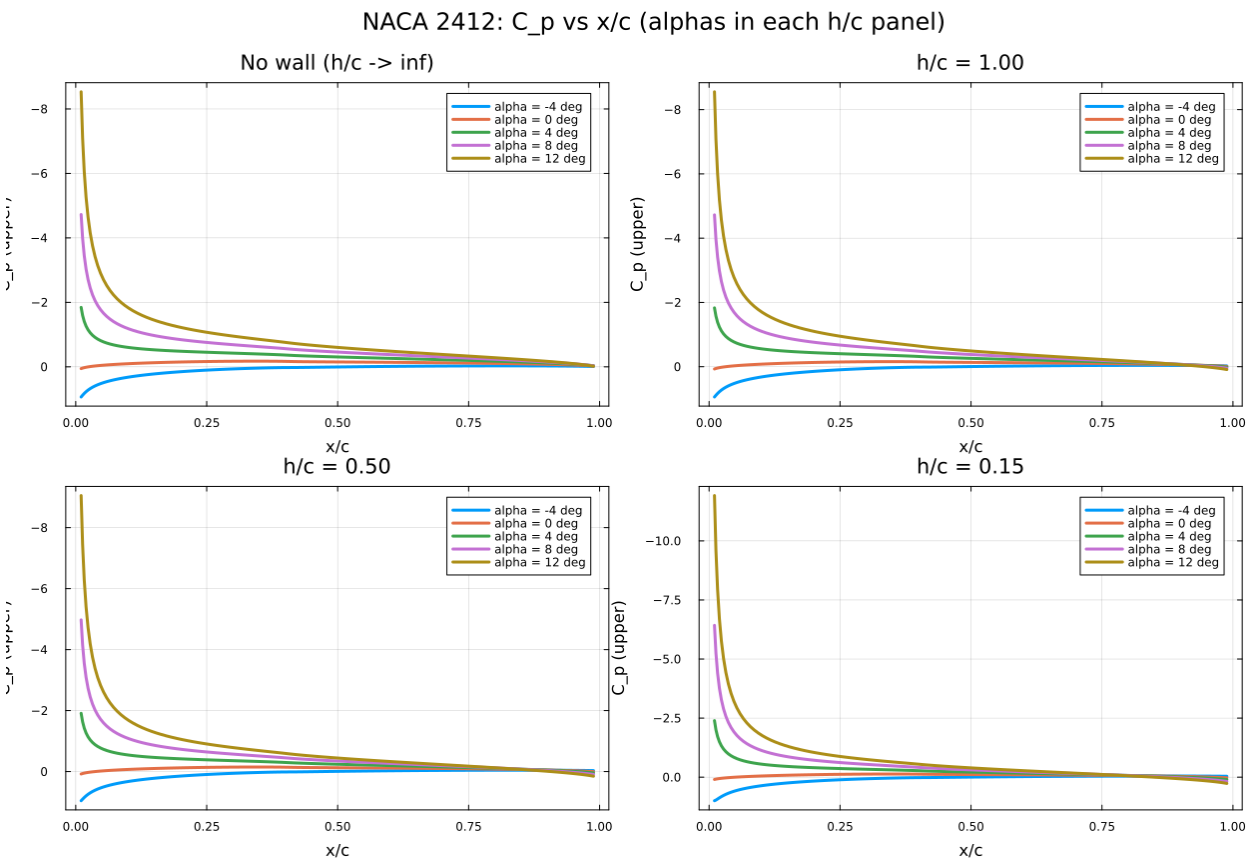

Figure 3: NACA 2412: C_p vs. x/c, grouped by h/c, with all requested α curves in each panel.

3.3 γ vs x/c

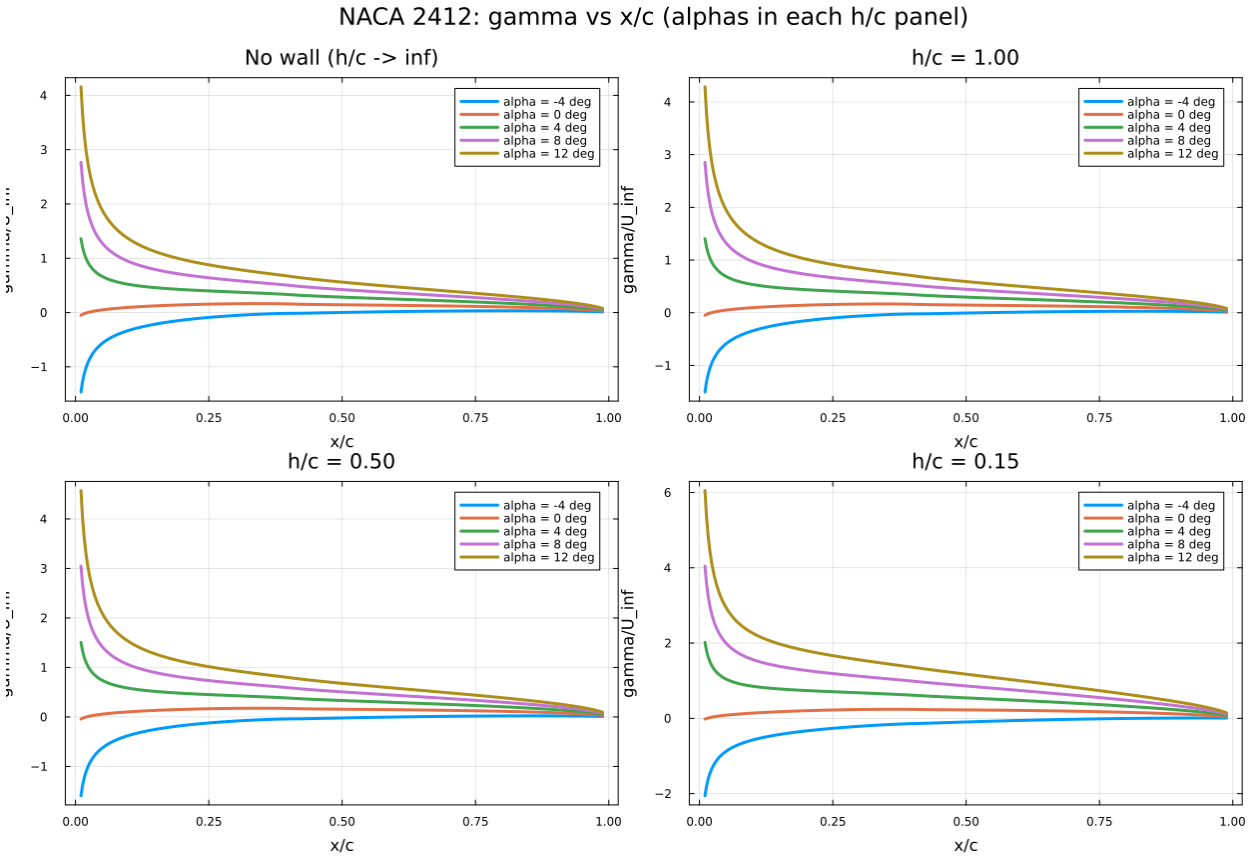

Figure 4: NACA 2412: γ distribution vs. x/c, grouped by h/c.

4 Verification Against Literature Reference

Figure 5 compares model C_p and XFOIL C_p for the no-ground case.

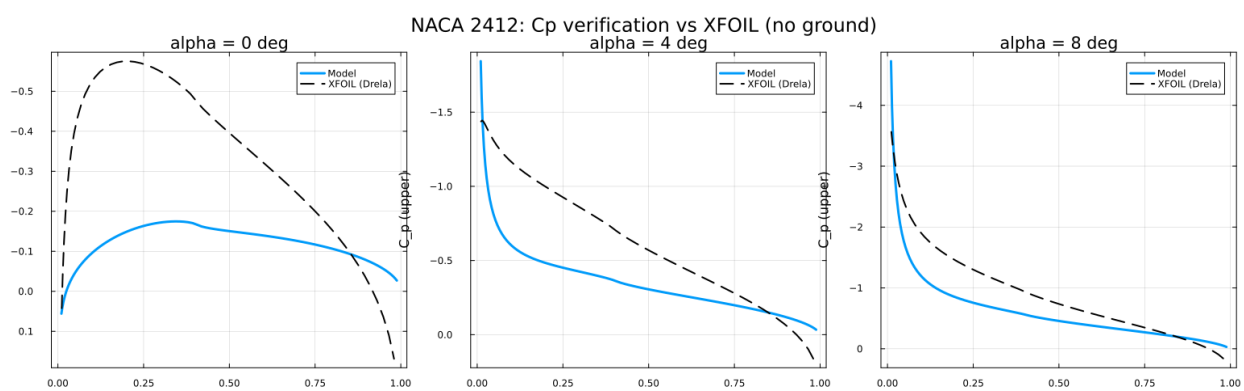

Figure 5: Verification plot: model C_p vs. XFOIL C_p (NACA 2412, no ground).

Table 2: Per-angle verification metrics (no-ground case).

α	C_p RMSE	$C_{L,\text{model}}$	$C_{L,\text{XFOIL}}$	ΔC_L
0°	0.2719	0.2277	0.2591	-0.0314
4°	0.3326	0.6655	0.7409	-0.0754
8°	0.4117	1.1001	1.2192	-0.1191

Interesting Point 1: Where the C_p difference is largest and smallest

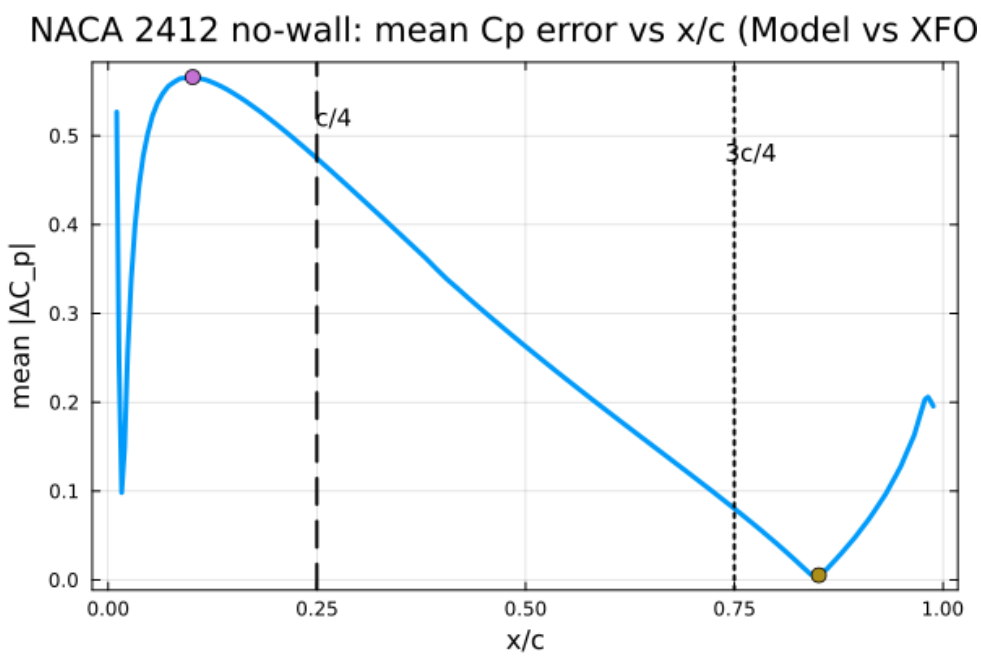

Figure 6: Mean absolute C_p difference along chord (no-ground, averaged over $\alpha = 0°, 4°, 8°$).

Using the no-ground verification set:

- Mean max error occurs at $x/c \approx 0.101$,
- Mean min error occurs at $x/c \approx 0.851$,
- $|\Delta C_p|(c/4) = 0.4746$,
- $|\Delta C_p|(3c/4) = 0.0804$.

I observed that the model–XFOIL divergence is much larger near quarter-chord than near three-quarter-chord. To me, this makes physical sense: in a thin-airfoil/camber-line model, the front part of the chord is where thickness effects and leading-edge suction peak sensitivity are strongest, so errors are amplified there. Farther downstream, the pressure gradients are milder, so the mismatch reduces.

Interesting Point 2: How accuracy changes with panel count

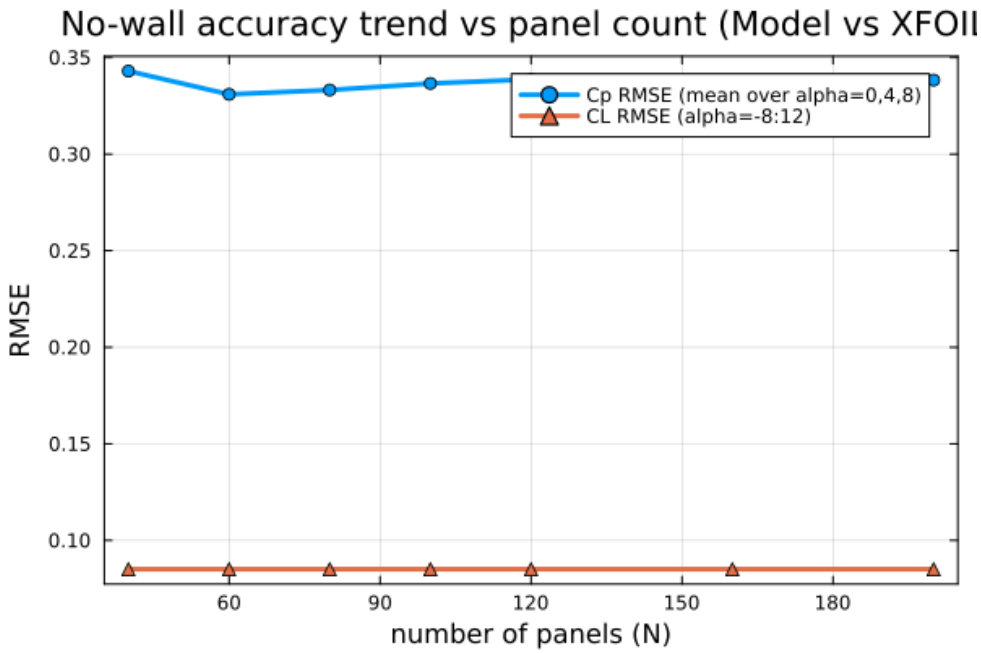

Figure 7: No-ground accuracy trend vs panel count for model vs XFOIL.

Table 3: Panel-count trend (no-ground): C_p RMSE and C_L RMSE.

N	C_p RMSE	C_L RMSE ($\alpha = -8^\circ : 12^\circ$)
40	0.3430	0.0851
60	0.3310	0.0851
80	0.3332	0.0851
100	0.3366	0.0851
120	0.3387	0.0851
160	0.3357	0.0851
200	0.3383	0.0851

Interpretation:

- Increasing panel count gives only small improvement after about $N \approx 60$.
- So the main error source is not mesh density; it is model-form simplification (camber-line thin-airfoil representation).
- XFOIL is more accurate because it solves flow on the full airfoil surface (with thickness) and uses a stronger aerodynamic formulation.

5 Discussion

1. C_L increases with α for all heights.

2. As h/c decreases, the C_L-α slope increases.
3. The strongest ground effect appears as we move the airfoil closer to the ground, which is also what we would intuitively expect.
4. C_p changes are strongest near the leading edge.
5. The verification trend matches shape and ordering from XFOIL, but differences are expected because this model is thin-airfoil based (camber-line, inviscid, no viscous effects).

6 Model Limits

- Inviscid model (no boundary layer, no separation, no stall).
- 2D only (no finite-span effects).
- Camber-line approximation is lower fidelity than full surface panel methods.

7 Conclusion

The procedure captures the expected ground-effect trend: smaller h/c gives stronger lift response and stronger leading-edge suction behavior. The requested C_p layout is satisfied, and a verification plot against the published XFOIL method is included.

Reference

M. Drela, "XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils," in *Low Reynolds Number Aerodynamics*, Springer, Lecture Notes in Engineering 54, 1989.